

Lab 1

Create VPC

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances. Mouse over a resource to highlight the related resources.

VPC settings

Resources to create

Create only the VPC resource or the VPC and other networking resources.

☐ VPC only ☒ VPC and more

Name tag auto-generation

Enter a value for the Name tag. This value will be used to auto-generate Name tags for all resources in the VPC.

☒ Auto-generate

ITI-46

IPv4 CIDR block

Determine the starting IP and the size of your VPC using CIDR notation.

10.0.0.0/16 65,536 IPs

CIDR block size must be between /16 and /28.

IPv6 CIDR block

☒ No IPv6 CIDR block
☐ Amazon-provided IPv6 CIDR block

Tenancy

Default

Preview

VPC

Your AWS virtual network

ITI-46-vpc

Subnets (4)

Subnets within this VPC

us-east-1a

ITI-46-subnet-public1-us-east-1a

ITI-46-subnet-private1-us-east-1a

us-east-1b

ITI-46-subnet-public2-us-east-1b

ITI-46-subnet-private2-us-east-1b

Route tables (3)

Route network traffic to resources

ITI-46-rtb-public

ITI-46-rtb-private1-us-east-1a

ITI-46-rtb-private2-us-east-1b

Your VPCs

VPCs VPC encryption controls

Your VPCs (2)

Last updated 1 minute ago

Actions

Create VPC

Find VPCs by attribute or tag

☐	Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv4 CIDR	IPv6 CIDR	DHCP option set
☐	-	vpc-00cbc00d1119ec001	Available	-	-	Off	172.31.0.0/16	-	dopt-0606235a6180cb...
☐	ITI-46-vpc	vpc-03eb2ee348d5de236	Available	-	-	Off	10.0.0.0/16	-	dopt-0606235a6180cb...

Subnets (10)

Last updated 1 minute ago

Actions

Create subnet

Find subnets by attribute or tag

☐	Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR	IPv6 CIDR	IPv6 CIDR association ID	Availa
☐	-	subnet-0d1316aef9304ff6e	Available	vpc-00cbc00d1119ec001	Off	172.31.0.0/20	-	-	4091
☐	-	subnet-0dc07c28589ace385	Available	vpc-00cbc00d1119ec001	Off	172.31.48.0/20	-	-	4091
☐	-	subnet-0aee8251845848a18	Available	vpc-00cbc00d1119ec001	Off	172.31.16.0/20	-	-	4091
☐	-	subnet-0bbb8f5b2cbe7e641	Available	vpc-00cbc00d1119ec001	Off	172.31.80.0/20	-	-	4091
☐	-	subnet-06c15b839ceae6a8e	Available	vpc-00cbc00d1119ec001	Off	172.31.64.0/20	-	-	4091
☐	-	subnet-06127d7b49e98ec2a	Available	vpc-00cbc00d1119ec001	Off	172.31.32.0/20	-	-	4091
☐	ITI-46-subnet-private1-us-east-1a	subnet-00138d934b23c6b15	Available	vpc-03eb2ee348d5de236 ITI...	Off	10.0.128.0/20	-	-	4091
☐	ITI-46-subnet-public1-us-east-1a	subnet-020a8e2ccc27f4862	Available	vpc-03eb2ee348d5de236 ITI...	Off	10.0.0.0/20	-	-	4091
☐	ITI-46-subnet-private2-us-east-1b	subnet-0db284e12ff5f49f7	Available	vpc-03eb2ee348d5de236 ITI...	Off	10.0.144.0/20	-	-	4091
☐	ITI-46-subnet-public2-us-east-1b	subnet-0dd0bdcc63346572c	Available	vpc-03eb2ee348d5de236 ITI...	Off	10.0.16.0/20	-	-	4091

rtb-0b2309ac14c2548b4 / ITI-46-rtb-public

Actions

Details

Route table ID

rtb-0b2309ac14c2548b4

VPC

vpc-03eb2ee348d5de236 | ITI-46-vpc

Main

No

Owner ID

557583209481

Explicit subnet associations

2 subnets

Edge associations

-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (2)

Filter routes

Destination	Target	Status	Propagated	Route O...
0.0.0.0/0	lgw-01d795ecd51bc5714	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Rou...

aws

Search

[Alt+S]

EC2

Dashboard

EC2 Global View

Events

Instances

- Instances
- Instance Types
- Launch Templates
- Spot Requests
- Savings Plans
- Reserved Instances
- Dedicated Hosts
- Capacity Reservations
- Capacity Manager

Images

- AMIs
- AMI Catalog

Elastic Block Store

- Volumes
- Snapshots
- Lifecycle Manager

Network & Security

- Security Groups
- Elastic IPs
- Placement Groups
- Key Pairs

Resources

You are using the following Amazon EC2 resources in the United States (N. Virginia) Region:

Instances (running)0

Auto Scaling Groups0

Capacity Reservations0 API Error

Dedicated Hosts0 API Error

Elastic IPs0

Instances0

Key pairs0

Load balancers0

Placement groups0

Security groups2

Snapshots0

Volumes0

Launch instance

To get started, launch an Amazon EC2 instance, which is a virtual server in the cloud.

Launch instanceMigrate a server

Note: Your instances will launch in the United States (N. Virginia) Region

Instance alarms

0 in alarm0 OK0 insufficient data

View in CloudWatch

Instances in alarm

Scheduled events

United States (N. Virginia)

No scheduled events

Service health

AWS Health Dashboard

An error occurred

An error occurred retrieving service health information

Diagnose with Amazon Q

Zones

Zone name	Zone ID
us-east-1a	use1-az6
us-east-1b	use1-az1
us-east-1c	use1-az2
us-east-1d	use1-az4
us-east-1e	use1-az3
us-east-1f	use1-az5

Launch an instance

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Name

web-server

Add additional tags

Application and OS Images (Amazon Machine Image)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Quick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI
ami-07ff62358b87c7116 (64-bit (x86), uefi-preferred) / ami-059afa9e3a9c7af0c (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.10.20260105.0 x86_64 HVM kernel-6.1

Architecture	Boot mode	AMI ID	Publish Date	Username
64-bit (x86)	uefi-preferred	ami-07ff62358b87c7116	2026-01-02	ec2-user

Verified provider

▼ Instance type [Info](#) | [Get advice](#)

Instance type

t3.micro

Family: t3 2 vCPU 1 GiB Memory Current generation: true

On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour On-Demand SUSE base pricing: 0.0104 USD per Hour

On-Demand Linux base pricing: 0.0104 USD per Hour On-Demand RHEL base pricing: 0.0392 USD per Hour

On-Demand Windows base pricing: 0.0196 USD per Hour

Free tier eligible

▼

☐ All generations

[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

ITI-test ▼

↻ [Create new key pair](#)

▼ Network settings [Info](#)

VPC - required [Info](#)

vpc-03eb2ee348d5de236 (ITI-46-vpc) 10.0.0.0/16 ▼

↻

Subnet [Info](#)

subnet-020a8e2ccc27f4862 ITI-46-subnet-public1-us-east-1a

VPC: vpc-03eb2ee348d5de236 Owner: 557583209481 Availability Zone: us-east-1a (use1-az6)

Zone type: Availability Zone IP addresses available: 4091 CIDR: 10.0.0.0/20 ▼

↻ [Create new subnet](#)

Auto-assign public IP [Info](#)

Enable ▼

Additional charges apply when outside of free tier allowance

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group

☐ Select existing security group

Security group name - required

web-SG

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters. Valid characters: a-z, A-Z, 0-9, spaces, and .-:/()#,@!+=&:~*!\$*

Description - required [Info](#)

launch-wizard-1 created 2026-01-16T18:30:35.776Z

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, 0.0.0.0/0, SSH) [Remove](#)

Type [Info](#)

ssh ▼

Protocol [Info](#)

TCP

Port range [Info](#)

22

Source type [Info](#)

Anywhere ▼

Source [Info](#)

Q Add CIDR, prefix list or security group

0.0.0.0/0 ✕

Description - optional [Info](#)

SSH

▼ Configure storage [Info](#)

[Advanced](#)

1x GiB ▼ Root volume, 3000 IOPS, Not encrypted

1x GiB ▼ EBS volume, 3000 IOPS, Not encrypted

[Remove](#)

[i](#) Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage

[×](#)[Add new volume](#)

[i](#) Click refresh to view backup information

The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

[↺](#)

0 x File systems

[Edit](#)

User data - optional [Info](#)

Upload a file with your user data or enter it in the field.

[↶ Choose file](#)

```
#!/bin/bash
yum update -y
yum install httpd -y
systemctl start httpd
systemctl enable httpd
echo "<!DOCTYPE html>
<html>
<head>
<style>
h1 {
  color: red;
  text-align: center;
  font-family: Arial, sans-serif;
  font-size: 50px;
}
```

☐ User data has already been base64 encoded

▼ Summary

Number of instances | Info

1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.10....read more
ami-07ff62358b87c7116

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
2 volume(s) - 16 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet. Data transfer charges are not included as part of the free tier allowance.

Cancel

Launch instance

Preview code

Instances (1) Info

Last updated less than a minute ago

Connect

Instance state ▼

Actions ▼

Launch instances ▼

Find Instance by attribute or tag (case-sensitive)

Running ▼

☐	Name	Instance ID	Instance state	Instance type	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP	IPv6 IPs	Monit
☐	web-server	i-068dc5d65cd95f979	Running	t3.micro	View alarms +	us-east-1a	ec2-18-215-255-151.co...	18.215.255.151	-	-	disabl

Instance summary for i-068dc5d65cd95f979 (web-server) Info

Last updated less than a minute ago

Connect

Instance state ▼

Actions ▼

Instance ID i-068dc5d65cd95f979	Public IPv4 address 18.215.255.151 open address	Private IPv4 addresses 10.0.0.175
IPv6 address -	Instance state Running	Public DNS ec2-18-215-255-151.compute-1.amazonaws.com open address
Hostname type IP name: ip-10-0-0-175.ec2.internal	Private IP DNS name (IPv4 only) ip-10-0-0-175.ec2.internal	Elastic IP addresses -
Answer private resource DNS name -	Instance type t3.micro	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations. Learn more
Auto-assigned IP address 18.215.255.151 [Public IP]	VPC ID vpc-03eb2ee348d5de236 (ITI-46-vpc)	Auto Scaling Group name -
IAM Role -	Subnet ID subnet-020a8e2ccc27f4862 (ITI-46-subnet-public1-us-east-1a)	Managed false
IMDSv2 Required	Instance ARN arn:aws:ec2:us-east-1:557583209481:instance/i-068dc5d65cd95f979	
Operator -		



Welcome to my first Web App 🚀

ITI Intake 46



Lab 2

Create bucket [Info](#)

Buckets are containers for data stored in S3.

General configuration

AWS Region

US East (N. Virginia) us-east-1

Bucket type [Info](#)

☒ General purpose

Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

☐ Directory

Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket name [Info](#)

Marwan-311

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9, periods (.), and hyphens (-). [Learn more](#)

Copy settings from existing bucket - optional

Only the bucket settings in the following configuration are copied.

[Choose bucket](#)

Format: s3://bucket/prefix

Object Ownership [Info](#)

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who can specify access to objects.

Object Ownership

☒ ACLs disabled (recommended)

All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.

☐ ACLs enabled

Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

Object Ownership

Bucket owner enforced

Block Public Access settings for this bucket

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket and its objects is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying any of these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the individual settings below to suit your specific storage use cases. [Learn more](#)

☐ Block all public access

Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

☐ Block public access to buckets and objects granted through new access control lists (ACLs)

S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing permissions that allow public access to S3 resources using ACLs.

☐ Block public access to buckets and objects granted through any access control lists (ACLs)

S3 will ignore all ACLs that grant public access to buckets and objects.

☐ Block public access to buckets and objects granted through new public bucket or access point policies

S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.

☐ Block public and cross-account access to buckets and objects through any public bucket or access point policies

S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

Turning off block all public access might result in this bucket and the objects within becoming public

AWS recommends that you turn on block all public access, unless public access is required for specific and verified use cases such as static website hosting.

☒ I acknowledge that the current settings might result in this bucket and the objects within becoming public.

Bucket Versioning

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. [Learn more](#)

Bucket Versioning

☐ Disable

☒ Enable

Tags - optional

You can use bucket tags to analyze, manage and specify permissions for a bucket. [Learn more](#)

 You can use s3:ListTagsForResource, s3:TagResource, and s3:UntagResource APIs to manage tags on S3 general purpose buckets for access control in addition to cost allocation and resource organization. To ensure a seamless transition, please provide permissions to s3:ListTagsForResource, s3:TagResource, and s3:UntagResource actions. [Learn more](#)

No tags associated with this bucket.

[Add new tag](#)

You can add up to 50 tags.

Default encryption [Info](#)

Server-side encryption is automatically applied to new objects stored in this bucket.

Encryption type [Info](#)

Secure your objects with two separate layers of encryption. For details on pricing, see [DSSE-KMS pricing](#) on the [Storage](#) tab of the [Amazon S3 pricing page](#).

☒ Server-side encryption with Amazon S3 managed keys (SSE-S3)

☐ Server-side encryption with AWS Key Management Service keys (SSE-KMS)

☐ Dual-layer server-side encryption with AWS Key Management Service keys (DSSE-KMS)

Bucket Key

Using an S3 Bucket Key for SSE-KMS reduces encryption costs by lowering calls to AWS KMS. S3 Bucket Keys aren't supported for DSSE-KMS. [Learn more](#)

☐ Disable

☒ Enable

► Advanced settings

- ✔

Successfully created bucket "marwan-311"
To upload files and folders, or to configure additional bucket settings, choose [View details](#).

View details

✕
- ⚠

The specified bucket settings for Block Public Access were not applied
You need s3:PutBucketPublicAccessBlock permission to modify individual Block Public Access on this bucket. After you or your AWS admin has updated your IAM permissions to allow s3:PutBucketPublicAccessBlock, go to [Edit bucket settings for Block Public Access](#).

Diagnose with Amazon Q

✕
- ⚠

Insufficient permissions to apply Bucket Versioning
You need the s3:PutBucketVersioning permission to apply Bucket Versioning on this bucket. After you or your AWS admin has updated your Identity and Access Management (IAM) permissions to allow s3:PutBucketVersioning, go to [edit Bucket Versioning](#).

Diagnose with Amazon Q

✕

General purpose buckets **All AWS Regions** Directory buckets

General purpose buckets (1) [Info](#)

[Copy ARN](#) [Empty](#) [Delete](#) [Create bucket](#)

Buckets are containers for data stored in S3.

Name

marwan-311

AWS Region

US East (N. Virginia) us-east-1

Creation date

January 16, 2026, 21:15:32 (UTC+02:00)

Account snapshot [Info](#)

Updated daily

[View dashboard](#)

Storage Lens provides visibility into storage usage and activity trends.

External access summary [Info](#)

Updated daily

External access findings help you identify bucket permissions that allow public access or access from other AWS accounts.

Files and folders (1 total, 57.2 KB)

All files and folders in this table will be uploaded.

☐

Name

☐

enem3.jpg

▼

Folder

▼

Type

image/jpeg

▼

Size

57.2 KB

Destination [Info](#)

Destination
[s3://marwan-311](#)

► Destination details
Bucket settings that impact new objects stored in the specified destination.

► Permissions
Grant public access and access to other AWS accounts.

▼ Properties
Specify storage class, encryption settings, tags, and more.

Storage class [Info](#)

Amazon S3 offers a range of storage classes designed for different use cases. [Learn more](#) or see [Amazon S3 pricing](#)

	Storage class	Designed for	Bucket type	Availability Zones	Min storage duration	Min billable object size	Monitoring and auto-tiering fees	Retrieval fees
☒	Standard	Frequently accessed data (more than once a month) with milliseconds access	General purpose	≥ 3	-	-	-	-
☐	Intelligent-Tiering	Data with changing or unknown access patterns	General purpose	≥ 3	-	-	Per-object fees apply for objects ≥ 128 KB	-
☐	Standard-IA	Infrequently accessed data (once a month) with milliseconds access	General purpose	≥ 3	30 days	128 KB	-	Per-GB fees apply
☐	One Zone-IA	Recreatable, infrequently accessed data (once a month) with milliseconds access	General purpose or directory	1	30 days	128 KB	-	Per-GB fees apply
☐	Glacier Instant Retrieval	Long-lived archive data accessed once a quarter with instant retrieval in milliseconds	General purpose	≥ 3	90 days	128 KB	-	Per-GB fees apply
☐	Glacier Flexible Retrieval (formerly Glacier)	Long-lived archive data accessed once a year with retrieval of minutes to hours	General purpose	≥ 3	90 days	-	-	Per-GB fees apply
☐	Glacier Deep Archive	Long-lived archive data accessed less than once a year with retrieval of hours	General purpose	≥ 3	180 days	-	-	Per-GB fees apply
☐	Reduced redundancy	Noncritical, frequently accessed data with milliseconds access (not recommended as S3 Standard is more cost effective)	General purpose	≥ 3	-	-	-	Per-GB fees apply

Close

Destination
s3://marwan-311

Succeeded

✔ 1 file, 57.2 KB (100.00%)

Failed
0 files, 0 B (0%)

Configuration

Find by name

Name	Folder	Type	Size	Status	Error
enem3.jpg	-	image/jpeg	57.2 KB	✔ Succeeded	-

Info

Copy S3 URI

Download

[Open ↗](#)

Object actions ▼

Properties

Permissions

Versions

✔ S3 URI Copied

Owner
216fa76ed34c249fc318e96312d3713065fff21713605fba311cf1223fbd27cf

AWS Region
US East (N. Virginia) us-east-1

Last modified
January 16, 2026, 21:19:29 (UTC+02:00)

Size
57.2 KB

Type
jpg

Key
enem3.jpg

s3://marwan-311/enem3.jpg

Amazon Resource Name (ARN)
arn:aws:s3:::marwan-311/enem3.jpg

Entity tag (Etag)
94a25b0e61a952554ad389d60e9eb21f

Object URL
<https://marwan-311.s3.us-east-1.amazonaws.com/enem3.jpg>



Lab 3

Resources

[Refresh](#)

You are using the following Amazon RDS resources in the US East (N. Virginia) region (used/quota)

DB Instances (0/40)

Allocated storage (0 TB/100 TB)

Instances and storage include Neptune and DocumentDB. [Increase DB instances limit](#)

DB Clusters (0/40)

Reserved instances (0/40)

Snapshots (0)

Manual

DB Cluster (0/100)

DB Instance (0/100)

Automated

DB Cluster (0)

DB Instance (0)

Recent events (0)

Event subscriptions (0/20)

Parameter groups (6)

Default (6)

Custom (0/100)

Option groups (5)

Default (5)

Custom (0/20)

Subnet groups (0/50)

[Supported platforms](#) [VPC](#)

[Default network](#) vpc-00cbc00d1119ec001

Create a database

Amazon Relational Database Service (RDS) makes it easy to set up, operate, and scale a relational database in the cloud.

[Create a database](#)

You can use a backup from Amazon S3 to restore and create a new Aurora MySQL and MySQL database.

[Restore from S3](#)

Note: your DB instances will launch in the **US East (N. Virginia)** region

Service health

[View service health dashboard](#)

Current status

Details

✓ Amazon Relational Database Service (N. Virginia)

Service is operating normally

Create database [Info](#)

Choose a database creation method

☒ Full configuration

You set all of the configuration options, including ones for availability, security, backups, and maintenance.

☐ Easy create

Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

Engine options

Engine type [Info](#)

☐ Aurora (MySQL Compatible)



☐ Aurora (PostgreSQL Compatible)



☒ MySQL



☐ PostgreSQL



☐ MariaDB



☐ Oracle

ORACLE

☐ Microsoft SQL Server



☐ IBM Db2

IBM Db2

Edition

☒ MySQL Community

▼ Hide filters

- ☒ Show only versions that support the Multi-AZ DB cluster [Info](#)
Create a Multi-AZ DB cluster with one primary DB instance and two readable standby DB instances. Multi-AZ DB clusters provide up to 2x faster transaction commit latency and automatic failover in typically under 35 seconds.
- ☒ Show only versions that support the Amazon RDS Optimized Writes [Info](#)
Amazon RDS Optimized Writes improves write throughput by up to 2x at no additional cost.

Engine version

MySQL 8.0.43

- ☐ Enable RDS Extended Support [Info](#)
Amazon RDS Extended Support is a [paid offering](#). By selecting this option, you consent to being charged for this offering if you are running your database major version past the RDS end of standard support date for that version. Check the end of standard support date for your major version in the [RDS for MySQL documentation](#).

Templates

Choose a sample template to meet your use case.

☐ Production

Use defaults for high availability and fast, consistent performance.

☐ Dev/Test

This instance is intended for development use outside of a production environment.

☒ Free tier

Use RDS Free Tier to develop new applications, test existing applications, or gain hands-on experience with Amazon RDS. [Info](#)

Availability and durability

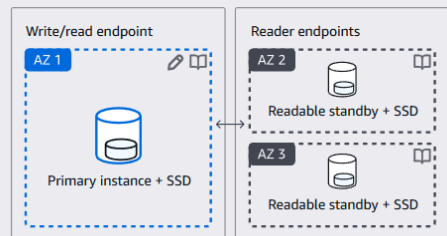
Availability and durability

Deployment options [Info](#)

Choose the deployment option that provides the availability and durability needed for your use case. AWS is committed to a certain level of uptime depending on the deployment option you choose. Learn more in the [Amazon RDS service level agreement \(SLA\)](#).

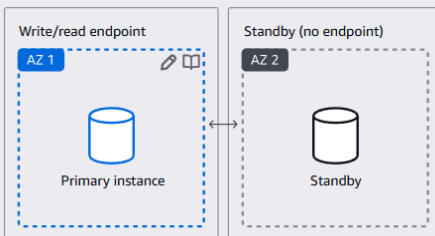
☐ Multi-AZ DB cluster deployment (3 instances)

- Creates a primary DB instance with two readable standbys in separate Availability Zones. This setup provides:
- 99.95% uptime
 - Redundancy across Availability Zones
 - Increased read capacity
 - Reduced write latency



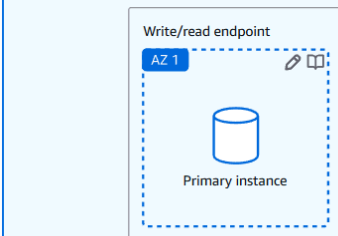
☐ Multi-AZ DB instance deployment (2 instances)

- Creates a primary DB instance with a non-readable standby instance in a separate Availability Zone. This setup provides:
- 99.95% uptime
 - Redundancy across Availability Zones



☒ Single-AZ DB instance deployment (1 instance)

- Creates a single DB instance without standby instances. This setup provides:
- 99.5% uptime
 - No data redundancy



Settings

DB instance identifier [Info](#)

Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.

marwan-311

The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 63 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

Credentials Settings

Master username [Info](#)

Type a login ID for the master user of your DB instance.

admin

1 to 16 alphanumeric characters. The first character must be a letter.

Credentials management

You can use AWS Secrets Manager or manage your master user credentials.

☐ Managed in AWS Secrets Manager - most secure

RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.

☒ Self managed

Create your own password or have RDS create a password that you manage.

☐ Auto generate password

Amazon RDS can generate a password for you, or you can specify your own password.

Master password [Info](#)

Password strength [Average](#)

Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / ' * @

Confirm master password [Info](#)

▼ Additional credentials settings

Database authentication options [Info](#)

- ☒ Password authentication
Authenticates using database passwords.
- ☐ Password and IAM database authentication
Authenticates using the database password and user credentials through AWS IAM users and roles.
- ☐ Password and Kerberos authentication
Choose a directory in which you want to allow authorized users to authenticate with this DB instance using Kerberos Authentication.

Instance configuration

The DB instance configuration options below are limited to those supported by the engine that you selected above.

DB instance class [Info](#)

▼ Hide filters

- ☐ Show instance classes that support Amazon RDS Optimized Writes [Info](#)
Amazon RDS Optimized Writes improves write throughput by up to 2x at no additional cost.

- ☐ Include previous generation classes

- ☐ Standard classes (includes m classes)
- ☐ Memory optimized classes (includes r and x classes)
- ☒ Burstable classes (includes t classes)

db.t4g.micro
2 vCPUs 1 GiB RAM EBS Bandwidth: Up to 2,085 Mbps Network: Up to 5 Gbps

Connectivity [Info](#)

Compute resource

Choose whether to set up a connection to a compute resource for this database. Setting up a connection will automatically change connectivity settings so that the compute resource can connect to this database.

- ☒ Don't connect to an EC2 compute resource
Don't set up a connection to a compute resource for this database. You can manually set up a connection to a compute resource later.
- ☐ Connect to an EC2 compute resource
Set up a connection to an EC2 compute resource for this database.

Virtual private cloud (VPC) [Info](#)

Choose the VPC. The VPC defines the virtual networking environment for this DB instance.

IT1-46-vpc (vpc-03eb2ee348d5de236)
4 Subnets, 2 Availability Zones

Only VPCs with a corresponding DB subnet group are listed.

- ① After a database is created, you can't change its VPC.

DB subnet group [Info](#)

Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

Create new DB Subnet Group

Public access [Info](#)

- ☒ Yes
RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.
- ☐ No
RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

VPC security group (firewall) [Info](#)

Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

- ☒ Choose existing
Choose existing VPC security groups
- ☐ Create new
Create new VPC security group

Public access [Info](#)

- ☒ Yes
RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.
- ☐ No
RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

VPC security group (firewall) [Info](#)

Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

- ☐ Choose existing
Choose existing VPC security groups
- ☒ Create new
Create new VPC security group

New VPC security group name

marwanragab

Availability Zone [Info](#)

No preference

RDS Proxy

RDS Proxy is a fully managed, highly available database proxy that improves application scalability, resiliency, and security.

- ☐ Create an RDS Proxy [Info](#)
RDS automatically creates an IAM role and a Secrets Manager secret for the proxy. RDS Proxy has additional costs. For more information, see [Amazon RDS Proxy pricing](#).

Certificate authority - optional [Info](#)

Using a server certificate provides an extra layer of security by validating that the connection is being made to an Amazon database. It does so by checking the server certificate that is automatically installed on all databases that you provision.

rds-ca-rsa2048-g1 (default)
Expiry: May 26, 2061

If you don't select a certificate authority, RDS chooses one for you.

▼ Additional configuration

Database port [Info](#)

TCP/IP port that the database will use for application connections.

3306

Creating database marwan-311

Your database might take a few minutes to launch. You can use settings from marwan-311 to simplify configuration of suggested database add-ons while we finish creating your DB for you.

Access denied to rds:DescribeGlobalClusters

You don't have permission to `rds:DescribeGlobalClusters`. To request access, copy the following text and send it to your AWS administrator. [Learn more about troubleshooting access denied errors.](#)

User: `arn:aws:iam::557583209481:user/kk_labs_user_249666`

Action: `rds:DescribeGlobalClusters`

On resource(s): `arn:aws:rds::557583209481:global-cluster:*`

Context: a service control policy explicitly denies the action

Connection details to your database marwan-311

This is the only time you can view this password. Copy and save the password for your reference. If you lose the password, you must modify your database to change it. You can use a SQL client application or utility to connect to your database.

[Learn about connecting to your database](#)

Master username

admin

Master password

Close

databases (1)

Filter by databases

DB identifier

Status

marwan-311

Creating

Size

Recommendations

CPU

db.t4g.micro

-

MySQL Workbench

File Edit View Database Tools Scripting Help

Welcome to MySQL Workbench

MySQL Workbench is the official graphical user interface (GUI) tool for MySQL. It allows you to design, create and browse your database schemas, work with database objects and insert data as well as

Setup New Connection

Connection Name:

Connection Method: Standard (TCP/IP)

Parameters

Hostname: 127.0.0.1

Username: admin

Password:

Default Schema:

Store Password For Connection

Please enter password for the following service:

Service: MySQL@127.0.0.1:3306

User: admin

Password:

Configure Server Management...

Test Connection Cancel OK

Successfully created database marwan-311

You can use settings from marwan-311 to simplify configuration of suggested database add-ons while we finish creating your DB for you.

[View connection details](#)

Access denied to rds:DescribeGlobalClusters

You don't have permission to `rds:DescribeGlobalClusters`. To request access, copy the following text and send it to your AWS administrator. [Learn more about troubleshooting access denied errors.](#)

User: `arn:aws:iam::557583209481:user/kk_labs_user_249666`

Action: `rds:DescribeGlobalClusters`

On resource(s): `arn:aws:rds::557583209481:global-cluster:*`

Context: a service control policy explicitly denies the action

Databases (1)

Filter by databases

DB identifier

Status

Role

Engine

Upgrade rollout order

Region ...

Size

Recommendations

CPU

Current activity

Ma

marwan-311

Config...

Instance

MySQL Co...

SECOND

us-east-1b

db.t4g.micro

-

no

Group resources

Modify

Actions

Create database